
Confidently Wrong: Calibration Risks and Safe Abstention in LLMs for African Languages

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With Apart Research

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Abstract

Large language models (LLMs) are increasingly deployed in African-language settings, but their reliability in those languages is largely unmeasured. We audit four open-weight LLMs—Qwen3-4B-Instruct, Gemma-4-12B, Aya-Expansive-8B, and AfroLlama-V1—on two African-language benchmarks, Uhura-TruthfulQA (truthfulness) and AfriMMLU (knowledge), scoring each answer option by its length-normalised log-probability. Among three general-purpose multilingual models (Qwen3, Gemma, Aya-Expansive), ECE in African languages exceeds English ECE in 97% of per-language comparisons ($p < 10^{-17}$, sign test). By contrast, AfroLlama-V1—trained specifically on six African languages—shows the *opposite* pattern: its ECE is *lower* in African languages than in English for 74% of comparisons, demonstrating that calibration gaps are not inherent to African-language evaluation but a consequence of general-purpose training distributions. We evaluate three post-hoc mitigations: per-language temperature scaling consistently reduces ECE to ≈ 0.03 across all models; few-shot prompting improves calibration for three of four models but causes systematic degradation in Gemma-4-12B, which we analyse in detail; selective abstention improves effective accuracy at the cost of coverage. Taken together, our results identify calibration gaps in general-purpose multilingual models as a measurable, necessary precondition for confident-misinformation risk in low-resource language deployment, and show that cheap post-hoc recalibration is an immediately deployable safety lever.

1 Introduction

LLMs are entering high-stakes, information-seeking use in African languages—health, legal, educational, and civic information—often where users cannot easily verify an answer. The safety failure we study is being *confidently wrong*: an incorrect answer delivered with high apparent confidence is more harmful than a hedged one, because users and downstream systems act on the stated confidence. A calibrated model, whose confidence tracks its accuracy, can instead *abstain*, declining or deferring when uncertain [11]. If calibration holds in English but breaks in African languages, the populations least served by current models are most exposed to confident misinformation. We test whether this gap exists, how large it is, and whether it can be cheaply mitigated.

We do not claim to have demonstrated downstream harm in a causal sense—doing so would require a user study or task-embedded evaluation beyond our scope—but we argue that the calibration gaps we measure are a *necessary precondition* for such harm: a system cannot cause confident-misinformation harm unless it is miscalibrated in the relevant language. Our results show that general-purpose multilingual models currently meet this precondition for the vast

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majority of African languages.

Our main contributions are:

1. An open-weight, reproducible calibration audit of four LLMs on two African-language benchmarks, using length-normalised log-probability scoring that is robust to instruction-tuned models.
2. A statistical demonstration (sign test, $p < 10^{-17}$) that calibration degrades from English to African languages in general-purpose multilingual models, combined with the novel finding that AfroLlama-V1—an African-language-specific model—shows the opposite pattern.
3. An evaluation of three post-hoc mitigations (temperature scaling, selective abstention, few-shot prompting), characterising when each helps and when it fails (Gemma-4-12B under few-shot prompting).
4. A deployment framework combining recalibration with confidence-thresholded abstention for safer real-world use.

2 Related Work

Modern neural networks are systematically overconfident; temperature scaling is the standard single-parameter post-hoc fix [6], and ECE is the canonical metric [9]. Calibration degrades under distribution shift [8], motivating our English-to-African analysis. For LLMs, log-probability confidence is more reliable than verbalized confidence [12], and abstention on uncertain outputs reduces hallucination and improves safety [11; 7].

Our benchmarks are Uhura [3] (truthfulness in low-resource African languages) and AfriMMLU within IrokoBench [1] (knowledge across 18 languages). AfroLlama-V1 [4] and Aya-Expanse [2] are the African-focused models we evaluate. To our knowledge, no prior work has audited calibration specifically in African languages, compared post-hoc mitigations across models, or identified the calibration reversal we observe in language-specialised models.

3 Methods

3.1 Models

We evaluate four open-weight models spanning size, training regime, and multilingual focus:

- **Qwen3-4B-Instruct** (4B parameters) [10]: a general-purpose instruction-tuned model with broad multilingual coverage.
- **Gemma-4-12B** (12B parameters) [5]: a multimodal model evaluated here in text-only mode; instruction-tuned.
- **Aya-Expanse-8B** (8B parameters) [2]: a multilingual model from Cohere explicitly designed for cross-lingual tasks.
- **AfroLlama-V1** (7B parameters, LLaMA-2 base) [4]: fine-tuned on six languages—Swahili, Zulu, Yoruba, Hausa, Xhosa, and English. Amharic and N. Sotho, which appear in our benchmarks, are *out-of-distribution* (OOD) for this model and are marked † throughout. We choose AfroLlama rather than its vanilla LLaMA-2-7B base because our central question is what African-language fine-tuning does to calibration; comparing AfroLlama against the three general-purpose models therefore directly operationalises the effect of African-language training on a fixed base architecture.

All models are loaded in `bfloat16` with `sdpa` attention, evaluated at temperature 0 (greedy decoding). Full HuggingFace IDs are in section A.

Software and compute. All experiments ran on a single NVIDIA Tesla A100 (40 GB) using PyTorch ≥ 2.5 , Transformers ≥ 4.51 , Accelerate ≥ 0.33 , and Python 3.10+. Models were loaded in `bf16` with scaled dot-product attention (`sdpa`), a 2048-token context limit, and greedy decoding (temperature 0). Temperature calibration and bootstrap resampling both use a fixed seed of 1234. Exact package versions and the full experiment configuration are archived in the repository.

3.2 Datasets

We use Uhura-TruthfulQA [3] (English and six African languages; truthfulness; variable number of candidates per question) and AfriMMLU [1] (18 languages; knowledge; four options per question). Because their language sets and tasks differ, we analyse them separately and never average across them. We use the term “African” to denote all languages other than English and French.

3.3 Scoring

For both datasets we score each option o with tokens $o_1, \dots, o_L$ given context x by its length-normalised log-probability:

$$s(o) = \frac{1}{L} \sum_{t=1}^L \log p(o_t \mid x, o_{<t}), \quad p(o) = \text{softmax}_o[s(o)], \quad \hat{y} = \arg \max_o p(o).$$

We deliberately avoid letter-logit scoring (reading the next-token probability of “A”/“B”/“C”/“D”): without this, Gemma-4 returns “A” for 75% of items, scoring below chance. Answer-text cloze scoring collapsed for no model.

3.4 Metrics

We report: expected calibration error (ECE, 15 equal-width bins); overconfidence (mean confidence minus accuracy); Brier score; error-detection AUROC; and AUARC, the area under the accuracy-rejection curve (full definitions in section B).

3.5 Interventions

Temperature scaling. We fit one temperature T^* per language by minimising ECE on a seeded 50% calibration split, evaluated on the held-out half, searching

$$T \in \{0.5, 0.6, 0.7, 0.8, 0.9, 1.0, 1.2, 1.5, 2.0, 3.0, 5.0\}.$$

Selective abstention. The model answers only when its calibrated confidence exceeds a per-language threshold chosen from the risk-coverage curve. We report AUARC integrated over all coverage levels.

Few-shot prompting. Five in-context examples per language are drawn from held-out items, formatted identically to the evaluation prompt, and prepended to each query.

4 Results

Table 1 summarises all models on both datasets.

4.1 Calibration degrades from English to African languages

Across all four models and both tasks, ECE and overconfidence are two to three times higher in African languages than English (tables 1, 4 and 5, fig. 2). Models are most overconfident precisely where they are least accurate.

Statistical summary. To assess whether this pattern is systematic, we apply a sign test across all per-language comparisons. For each model-dataset pair we count how many African

Table 1: English versus African-language performance by model and dataset (zero-shot). **Acc_{en}** and **Acc_{afr}** are accuracy on English and African-language splits (Acc_{afr} is a macro-average over all African languages). **ECE_{en}** and **ECE_{afr}** are Expected Calibration Error ($\downarrow$); **ECE_{TS}** is ECE after per-language temperature scaling. **AUARC** measures abstention quality ($\uparrow$; random baseline $\approx$ Acc_{afr}). $\dagger$ AfroLlama on Amharic and N. Sotho is out-of-distribution. Bootstrap 95% CIs for ECE_{en}, ECE_{afr}, Acc_{en}, and Acc_{afr} are in table 7.

Model (params)	Dataset	Acc _{en}	Acc _{afr}	ECE _{en}	ECE _{afr}	ECE _{TS}	AUARC
Qwen3-4B-Instruct (4B)	Uhura-TQA	.381	.320	.072	.169	.030	.359
Gemma-4-12B (12B)	Uhura-TQA	.329	.342	.212	.257	.033	.387
Aya-Expanse-8B (8B)	Uhura-TQA	.358	.308	.075	.200	.043	.347
AfroLlama-V1 (7B) $\dagger$	Uhura-TQA	.292	.325	.177	.135	.036	.345
Qwen3-4B-Instruct (4B)	AfriMMLU	.526	.296	.062	.152	.021	.349
Gemma-4-12B (12B)	AfriMMLU	.296	.269	.311	.341	.075	.289
Aya-Expanse-8B (8B)	AfriMMLU	.520	.287	.057	.180	.021	.335
AfroLlama-V1 (7B) $\dagger$	AfriMMLU	.356	.279	.225	.203	.030	.310

languages have ECE exceeding the English baseline for the same model. Among the three general-purpose models (Qwen3, Aya-Expanse, Gemma), 67 of 69 per-language comparisons show ECE(African) > ECE(English) (97.1%). Under the null hypothesis that each comparison is equally likely to go either way ($p = 0.5$), this proportion is extremely unlikely ($p < 10^{-17}$, exact binomial test), confirming the gap is systematic rather than incidental. We note that observations within a model are not fully independent (shared weights, similar prompts), so this statistic should be interpreted as a summary measure rather than a fully rigorous inference.

4.2 Calibration reversal in an African-language specialist

AfroLlama-V1 shows a striking reversal: its ECE is *lower* in African languages than in English for 17 of 23 per-language comparisons (74%), the opposite of the general-purpose models. Crucially, this improvement is concentrated in AfroLlama’s six training languages—all five non-English training languages (Swahili, Zulu, Yoruba, Hausa, Xhosa) show lower ECE than English on both benchmarks. AfroLlama’s OOD languages (Amharic, N. Sotho on Uhura; the remaining AfriMMLU languages) show mixed results, consistent with distribution shift.

This reversal is not an artefact of lower accuracy trivially shrinking ECE: AfroLlama’s accuracy on its training languages (e.g. Yoruba .371, Hausa .363 on Uhura) is comparable to or above its English accuracy (.292), so the ECE improvement is genuine. **This finding has a practical implication: calibration gaps in African languages are not inherent properties of the languages themselves but a consequence of whether models have been trained on those languages.** Targeted fine-tuning appears to simultaneously improve accuracy and calibration alignment.

Per-language breakdowns (Acc / ECE for every language and model) are in section E. Several accuracy figures are only modestly above the 0.25 random baseline for 4-way MCQ, so the overconfidence we measure occurs precisely where users are most likely to receive a confidently wrong answer.

4.3 Mitigations

Temperature scaling. Per-language temperature scaling cuts pooled ECE to approximately 0.02–0.07 for every model (table 1, ECE_{TS}; fig. 3). Optimal temperatures range from 1.2 to 5.0 across languages (see section E), confirming that a single global temperature is suboptimal. Temperature scaling is the most effective, lowest-cost mitigation we evaluate—it requires only a small calibration split and a line search, with no retraining.

Selective abstention. Abstention helps but is limited by a weak uncertainty signal: AUROC is approximately 0.55 across all models, meaning the confidence score is only slightly better than random at ranking correct above incorrect answers. As a result, AUARC sits only modestly above the baseline of answering everything. The durable fix for weak abstention is a better calibrated or more discriminative uncertainty signal, which depends on improved African-language training data.

Few-shot prompting. Table 2 shows zero-shot versus 5-shot results on Uhura-TruthfulQA. Three models—Qwen3, AfroLlama, and Aya-Expanse—improve modestly in both accuracy (+0.016–0.026) and ECE (−0.011–−0.022). **Gemma-4-12B is a clear exception:** ECE rises from .251 to .332 (+0.081) and overconfidence increases by the same magnitude, while accuracy falls by −0.035. This degradation occurs in five of six languages.

We attribute Gemma’s behaviour to its RLHF tuning: when shown examples of direct, confident answers, the model inflates its softmax outputs to match the exemplar style without a corresponding improvement in answer quality [6]. Critically, temperature scaling remains effective even for 5-shot Gemma ($ECE_{TS} = .033$), confirming the miscalibration is systematic and remediable post-hoc. **Practical recommendation:** for RLHF-tuned models, few-shot prompting should be combined with post-hoc recalibration or validated on a held-out calibration split before deployment.

Table 2: Zero-shot vs. 5-shot calibration on Uhura-TruthfulQA (macro-averaged over languages). Δ values are signed differences (negative = improvement). **Gemma-4-12B shows calibration degradation under few-shot prompting** and is discussed in section 4.

Model	Acc _{0s}	Acc _{5s}	Δ Acc	ECE _{0s}	ECE _{5s}	Δ ECE
Qwen3-4B	.329	.355	+0.026	.155	.141	−0.013
AfroLlama-V1	.320	.339	+0.018	.141	.119	−0.022
Aya-Expanse-8B	.315	.331	+0.016	.182	.171	−0.011
Gemma-4-12B	.340	.305	−0.035	.251	.332	+0.081

5 Discussion and Limitations

Implications for African AI safety. In African languages these models are at once less accurate and more overconfident—the worst combination for trust—since confident wrong answers reach users precisely where independent verification is hardest. Encouragingly, much of the failure is an output-layer problem: cheap, post-hoc, per-language temperature scaling closes most of the calibration gap with no retraining, so safer deployment is feasible today. AfroLlama’s calibration reversal shows that targeted training is a durable path to genuine improvement in both accuracy and calibration alignment. We distil these findings into a deployment framework (section D, fig. 1): recalibrate first, then route on calibrated confidence to answer, request more context, or abstain and defer, logging low-confidence queries to drive African-language data collection.

ECE differences may partly reflect benchmark translation quality rather than model miscalibration, and tokenization inefficiency in African languages can inflate log-probability variance; a native-speaker item audit would be needed to separate these effects. Temperature estimates rest on ≈ 250 calibration examples per language with no cross-validation, so per-language ECE_{TS} figures carry non-negligible variance; bootstrap 95% CIs (table 7) quantify this for the zero-shot point estimates. Seven of 42 language–model pairs have $T^* = 5.0$, the upper boundary of our search grid; the true optimal temperature for those pairs may exceed 5.0, so ECE_{TS} for those cases is an upper bound on what temperature scaling can achieve. The few-shot experiment covers Uhura-TruthfulQA only, and we evaluate four open-weight models; reasoning models,

closed APIs, and broader topic coverage are left to future work.

6 Conclusion

Open LLMs are confidently wrong in African languages: less accurate and more overconfident than in English, consistently across four models and two tasks. Among general-purpose multilingual models, this calibration gap is highly systematic (97% of per-language comparisons, $p < 10^{-17}$). AfroLlama-V1, trained specifically on African languages, shows the opposite pattern, suggesting that calibration gaps are remediable through targeted training rather than being an inherent property of these languages.

Post-hoc per-language temperature scaling is a cheap, effective safety lever that reduces ECE to ≈ 0.03 for all models with no retraining required. Few-shot prompting helps most models but degrades Gemma-4-12B; it should be validated on a held-out calibration split before deployment. Selective abstention is currently limited by a weak uncertainty signal and should be viewed as a complement to, not a substitute for, recalibration.

We emphasise that this work measures a necessary precondition for harm rather than harm itself. Quantifying downstream consequences—in terms of user decisions, trust, or real-world outcomes—requires task-embedded evaluation with real users, which we leave to future work. The present contribution is to establish baseline measurements that make such evaluation possible, and to show that at least one remediation (temperature scaling) is immediately deployable.

Code and data. All code, predictions, figures, and tables are available in our GitHub repository. Datasets: Uhura-TruthfulQA and AfriMMLU.

LLM usage. We used Claude to help build and debug the evaluation pipeline and to draft this report; all results were produced by our code and independently verified.

References

- [1] D. I. Adelani et al. IrokoBench: A new benchmark for African languages in the age of large language models. *arXiv preprint arXiv:2406.03368*, 2024.
- [2] V. Aryabumi et al. Aya expanse: Combining research breakthroughs for a new multilingual frontier. *arXiv preprint arXiv:2412.04261*, 2024.
- [3] E. Bayes, I. A. Azime, J. O. Alabi, J. Kgomo, T. Eloundou, E. Proehl, K. Chen, I. Khadir, N. A. Etori, S. H. Muhammad, C. Mpanza, I. P. Thete, D. Klakow, and D. I. Adelani. Uhura: A benchmark for evaluating scientific question answering and truthfulness in low-resource African languages. *arXiv preprint arXiv:2412.00948*, 2024.
- [4] B. F. P. Dossou et al. AfroLlama-V1: LLaMA-2 fine-tuned on African languages, 2024. Model: `Jacaranda/AfroLlama_V1` on Hugging Face. 7B parameters (LLaMA-2 base). Training languages: Swahili, Zulu, Yoruba, Hausa, Xhosa, English. https://huggingface.co/Jacaranda/AfroLlama_V1.
- [5] Gemma Team et al. Gemma: Open models based on Gemini research and technology. *arXiv preprint arXiv:2403.08295*, 2024.
- [6] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger. On calibration of modern neural networks. In *Proceedings of the 34th International Conference on Machine Learning (ICML)*, pages 1321–1330. PMLR, 2017.
- [7] L. Kuhn, Y. Gal, and S. Farquhar. Semantic uncertainty: Linguistic invariances for uncertainty estimation in natural language generation. In *International Conference on Learning Representations (ICLR)*, 2023.
- [8] M. Minderer, J. Djolonga, R. Romijnders, F. Hubis, X. Zhai, N. Houlsby, D. Tran, and

- M. Lucic. Revisiting the calibration of modern neural networks. *Advances in Neural Information Processing Systems*, 34:15682–15694, 2021.
- [9] M. P. Naeni, G. Cooper, and M. Hauskrecht. Obtaining well calibrated probabilities using Bayesian binning into quantiles. In *Proceedings of the 29th AAAI Conference on Artificial Intelligence*, pages 2901–2907, 2015.
- [10] Qwen Team. Qwen3 technical report, 2025. Alibaba Cloud. <https://qwenlm.github.io/blog/qwen3/>.
- [11] C. Tomani, K. Chaudhuri, I. Evtimov, D. Cremers, and M. Ibrahim. Uncertainty-based abstention in LLMs improves safety and reduces hallucinations. *arXiv preprint arXiv:2404.10960*, 2024.
- [12] M. Xiong, Z. Hu, X. Lu, Y. Li, J. Fu, J. He, and B. Hooi. Can LLMs express their uncertainty? an empirical evaluation of confidence elicitation in LLMs. In *International Conference on Learning Representations (ICLR)*, 2024.

A Models

Table 3: Open-weight models evaluated.

Model	Params	Type	HuggingFace ID
Qwen3-4B-Instruct	4B	Instruction-tuned	Qwen/Qwen3-4B-Instruct
Gemma-4-12B	12B	Instruction-tuned	google/gemma-4-12b-it
Aya-Expansive-8B	8B	Multilingual	CohereLabs/aya-expansive-8b
AfroLlama-V1	7B	Africa-specific	Jacaranda/AfroLlama_V1

AfroLlama-V1 is based on LLaMA-2-7B and was fine-tuned on Swahili, Zulu, Yoruba, Hausa, Xhosa, and English. Amharic and N. Sotho (present in our benchmarks) are out-of-distribution for this model.

B Metric Definitions

For an item with options o and gold answer y , the cloze score of option o with tokens $o_1, \dots, o_L$ given context x is the length-normalised log-probability

$$s(o) = \frac{1}{L} \sum_{t=1}^L \log p(o_t \mid x, o_{<t}), \quad p(o) = \text{softmax}_o[s(o)], \quad \hat{y} = \arg \max_o p(o), \quad c = \max_o p(o).$$

Over N items with correctness $y_i \in \{0, 1\}$, confidence c_i , and $B = 15$ equal-width confidence bins $\{B_b\}$:

$$\text{ECE} = \sum_{b=1}^B \frac{|B_b|}{N} |\text{Acc}(B_b) - \text{conf}(B_b)|, \quad \text{Overconfidence} = \frac{1}{N} \sum_i c_i - \frac{1}{N} \sum_i y_i,$$

$$\text{Brier} = \frac{1}{N} \sum_i (c_i - y_i)^2.$$

AUROC is the area under the ROC curve using c to rank correct above incorrect items. AUARC is the area under the accuracy-rejection curve: items are answered most-confident-first and accuracy on the answered subset is integrated over coverage. Temperature scaling rescales logits by a single $T > 0$, $p_T(o) = \text{softmax}_o[s(o)/T]$; we fit $T^* = \arg \min_T \text{ECE}$ on a seeded 50% calibration split.

C Prompts

Each option is scored as a continuation of the prompt; the prompt lists no answer letters.

Uhura-TruthfulQA (truthfulness):

Q: {question}

A:

AfriMMLU (knowledge):

The following is a multiple-choice question about {subject}.

Give the correct answer.

Question: {question}

Answer:

For 5-shot prompting, the prompt is prefixed with five solved examples in the same language, each formatted as above followed by its gold answer text, separated by blank lines; examples are held out from evaluation.

D Deployment Framework

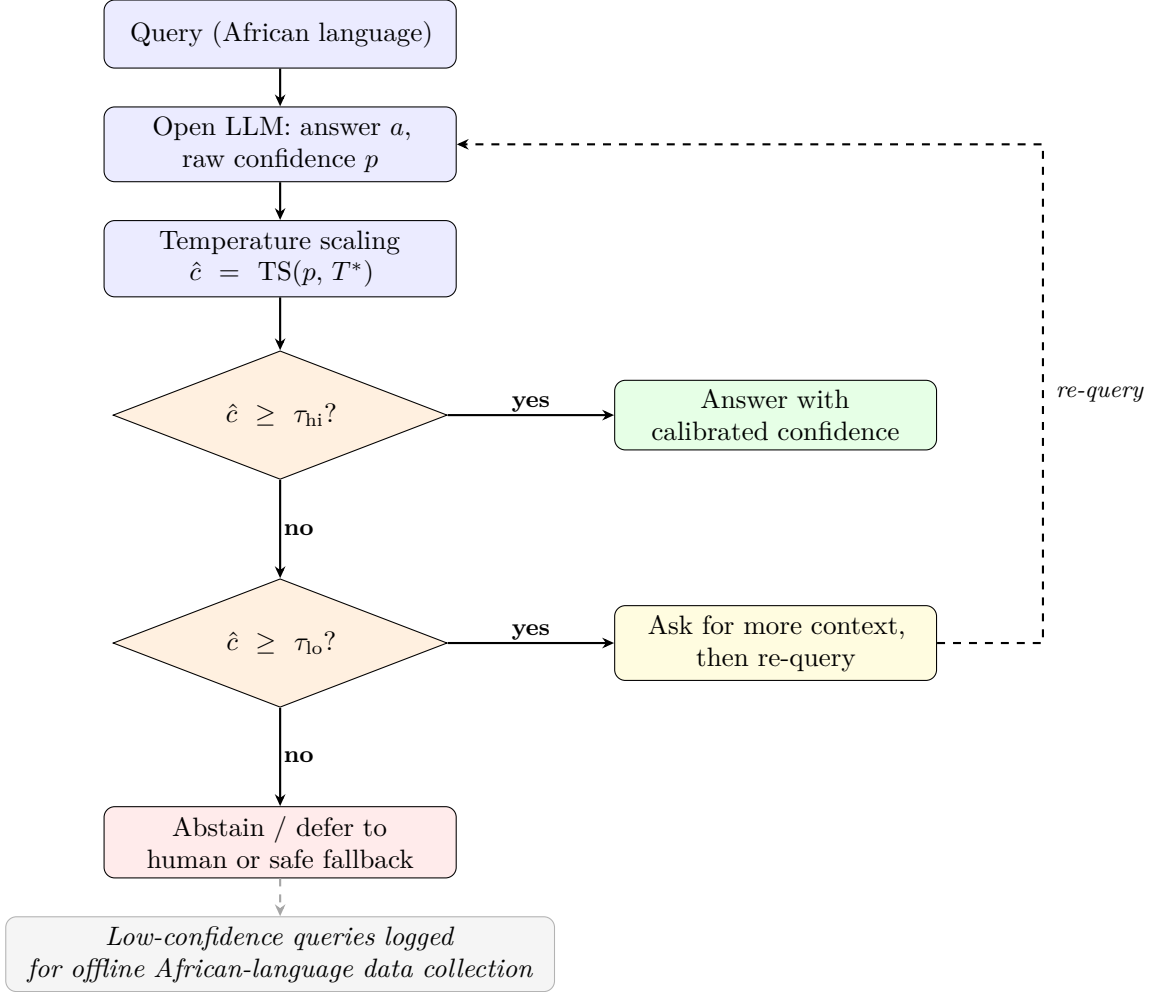


Figure 1: Deployment framework for LLMs in African languages. Recalibrate first with per-language temperature scaling, then route on calibrated confidence $\hat{c}$: high values ($\hat{c} \geq \tau_{hi}$) are answered with the stated confidence; medium values trigger a request for more context followed by re-query; low values ($\hat{c} < \tau_{lo}$) are abstained or deferred to a human or safe fallback. Thresholds τ are set per language from the risk-coverage curve. Low-confidence queries are logged to drive offline African-language data collection.

E Additional Results

Table 4: Per-language accuracy and ECE on Uhura-TruthfulQA (zero-shot). Each cell shows Acc / ECE. **Bold** ECE values exceed the English baseline for that model. † OOD for AfroLlama-V1.

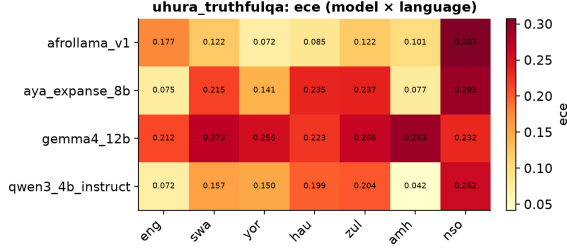
Language	AfroLlama-V1	Aya-Expanse-8B	Gemma-4-12B	Qwen3-4B
English (eng)	.292 / .177	.358 / .075	.329 / .212	.381 / .072
Amharic (amh) [†]	.294 / .101	.302 / .077	.274 / .293	.373 / .042
Hausa (hau)	.363 / .085	.302 / .235	.397 / .223	.308 / .199
N. Sotho (nso) [†]	.246 / .307	.260 / .293	.379 / .232	.270 / .262
Swahili (swa)	.341 / .122	.314 / .215	.328 / .273	.335 / .157
Yoruba (yor)	.371 / .072	.351 / .141	.329 / .256	.310 / .150
Zulu (zul)	.337 / .122	.319 / .237	.343 / .266	.324 / .204

Table 5: Per-language accuracy and ECE on AfriMMLU (zero-shot), representative subset of 17 African languages. Each cell shows Acc / ECE. **Bold** ECE values exceed the English baseline for that model.

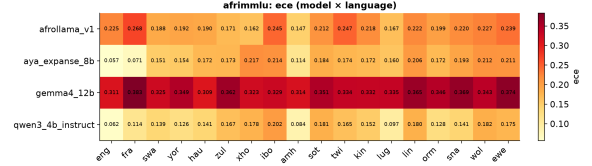
Language	AfroLlama-V1	Aya-Expanse-8B	Gemma-4-12B	Qwen3-4B
English (eng)	.356 / .225	.520 / .057	.296 / .311	.526 / .062
Amharic (amh)	.258 / .147	.240 / .114	.268 / .314	.286 / .084
Hausa (hau)	.294 / .190	.308 / .172	.308 / .309	.304 / .141
Swahili (swa)	.294 / .188	.312 / .151	.280 / .325	.312 / .139
Yoruba (yor)	.290 / .192	.300 / .154	.250 / .349	.308 / .126
Zulu (zul)	.310 / .171	.292 / .173	.252 / .362	.286 / .167
Ewe (ewe)	.244 / .239	.274 / .211	.252 / .374	.288 / .175
Igbo (ibo)	.252 / .245	.262 / .214	.288 / .329	.258 / .202
French (fra)	.280 / .268	.442 / .071	.244 / .383	.372 / .114

Table 6: Per-language temperature scaling on Uhura-TruthfulQA for AfroLlama-V1 and Qwen3-4B. T^* is selected on the calibration half; ECE values are on the held-out half.

Language	AfroLlama-V1			Qwen3-4B		
	T^*	ECE_{bef}	ECE_{aft}	T^*	ECE_{bef}	ECE_{aft}
English	5.0	.175	.069	1.5	.096	.074
Amharic	5.0	.115	.038	1.2	.056	.043
Hausa	2.0	.085	.077	5.0	.203	.027
N. Sotho	5.0	.307	.093	5.0	.260	.067
Swahili	2.0	.160	.074	3.0	.182	.043
Yoruba	1.5	.078	.064	5.0	.133	.062
Zulu	2.0	.126	.046	5.0	.191	.085
<i>Pooled</i>	–	.143	.036	–	.151	.030

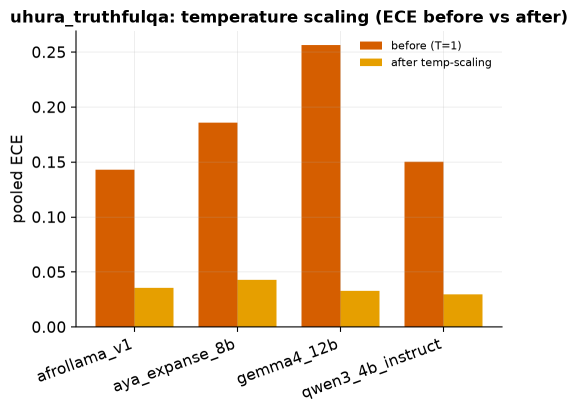


(a) Uhura-TruthfulQA

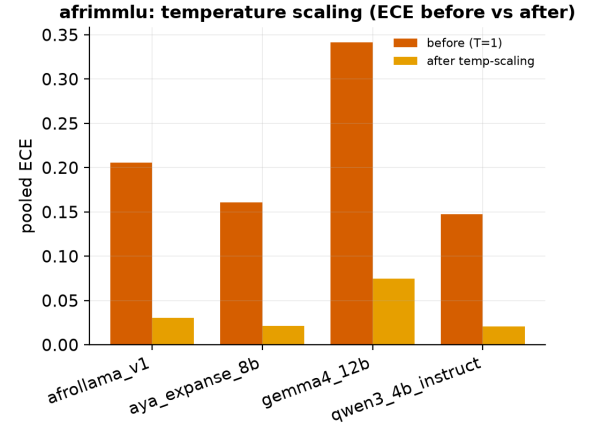


(b) AfriMMLU

Figure 2: ECE by model and language. ECE is low in English and rises across African languages for general-purpose models; AfroLlama-V1 shows the reverse pattern in its training languages.

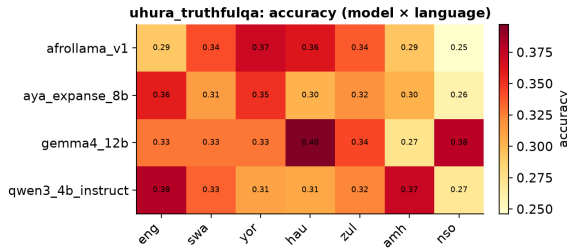


(a) Uhura-TruthfulQA

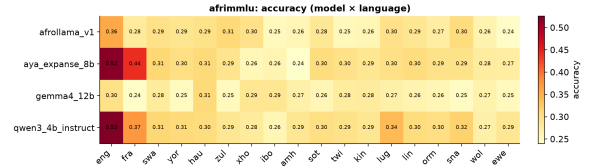


(b) AfriMMLU

Figure 3: Pooled ECE before and after per-language temperature scaling. ECE drops to ≈ 0.03 for all models on both datasets.



(a) Uhura-TruthfulQA



(b) AfriMMLU

Figure 4: Accuracy by model and language. Long English and French spokes collapse toward chance across African languages for general-purpose models.

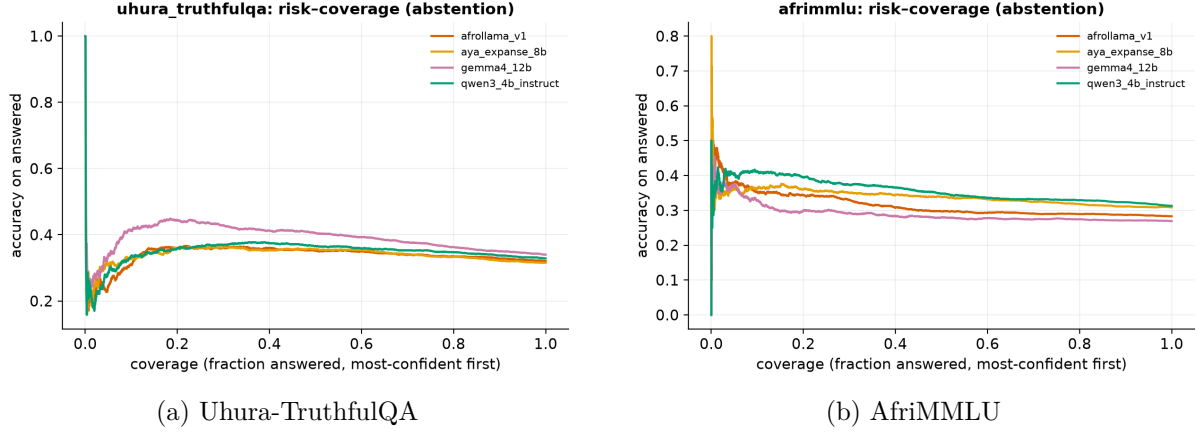


Figure 5: Accuracy-rejection curves under selective abstention. AUARC sits only modestly above baseline, reflecting a weak uncertainty signal (AUROC ≈ 0.55).

F Bootstrap Confidence Intervals

Table 7 reports 95% bootstrap confidence intervals (2,000 draws, stratified by language, seed 1234) for the four primary point estimates in table 1. Intervals are computed by resampling within each language stratum and macro-averaging ECE and accuracy across languages, matching the aggregation used in the main table.

Table 7: Bootstrap 95% confidence intervals for ECE and accuracy in table 1 (zero-shot, 2,000 bootstrap draws, stratified by language, seed 1234). Each cell shows point estimate [lower, upper].

Model	Dataset	ECE_{en}	ECE_{afr}	Acc_{en}	Acc_{afr}
Qwen3-4B	Uhura-TQA	.072 [.060, .113]	.169 [.160, .186]	.381 [.347, .413]	.320 [.306, .333]
Gemma-4-12B	Uhura-TQA	.212 [.182, .245]	.257 [.246, .273]	.329 [.297, .362]	.342 [.328, .355]
Aya-Expense-8B	Uhura-TQA	.075 [.062, .114]	.199 [.187, .214]	.358 [.326, .392]	.308 [.295, .321]
AfroLlama-V1	Uhura-TQA	.177 [.148, .212]	.135 [.126, .151]	.292 [.261, .325]	.325 [.312, .339]
Qwen3-4B	AfriMMLU	.062 [.052, .116]	.150 [.147, .165]	.526 [.482, .572]	.300 [.290, .310]
Gemma-4-12B	AfriMMLU	.311 [.273, .354]	.344 [.335, .355]	.296 [.258, .336]	.268 [.259, .277]
Aya-Expense-8B	AfriMMLU	.057 [.048, .113]	.173 [.170, .189]	.520 [.476, .564]	.296 [.286, .306]
AfroLlama-V1	AfriMMLU	.225 [.187, .267]	.207 [.201, .219]	.356 [.312, .398]	.279 [.269, .288]

The English ECE intervals are notably wider than the African-language intervals because English evaluation is a single-language stratum (no macro-averaging), while African ECE is macro-averaged over 5–17 languages, which reduces bootstrap variance. The key qualitative finding— ECE_{afr} consistently exceeds ECE_{en} for general-purpose models and is consistently *lower* for AfroLlama-V1 on Uhura-TQA—is preserved across the full width of all intervals: the CI for ECE_{afr} does not overlap with the CI for ECE_{en} in any general-purpose model row.